Logistic Regression (Loan Approval Prediction)

Assignment

PW SKILLS

Submission Requirements
Students must submit:

- Google Colab Notebook
- All code
- Exploratory Data Analysis
- Model evaluation metrics
- Regression equation
- Interpretation of coefficients

Problem Statement
You are working as a Data Analyst in a financial institution. The company wants to build a model to predict whether a loan application will be approved (1) or not approved (0) based on customer details.

You are given a dataset:
Dataset Link : Loan Approval Prediction

Your task is to perform Exploratory Data Analysis (EDA), build a Logistic Regression model, evaluate the model, and interpret the results.

Q1. Data Loading & Understanding
a) Load the dataset into Python.
b) Display the first 5 rows.
c) Check the shape of the dataset.
d) Display the data types of each column.

Q2. Data Cleaning
a) Check for missing values in the dataset.
b) Handle missing values appropriately.
c) Identify categorical columns present in the dataset.

Q3. Exploratory Data Analysis (EDA)
a) Plot the distribution of the target variable.
b) Analyze the relationship between one numerical feature and the target variable.
c) Analyze the relationship between Years of Employment and the target variable.

Q4. Outlier Detection and Treatment
a) Detect outliers in numerical columns using the IQR method.
b) Treat outliers using capping techniques

Q5. Convert target variables into numerical format suitable for model building and drop unnecessary columns.

Q6. Feature Selection and Data Splitting
a) Separate independent variables (X) and dependent variable (y).

b) Split the dataset into training and testing sets.

Q7. Apply feature scaling to the dataset using StandardScaler.

Q8. Logistic Regression Model Building
a) Train a Logistic Regression model.
b) Predict the output for test data.

Q9. Model Evaluation – Confusion Matrix
a) Generate the confusion matrix.
b) Interpret the results.

Q10. Model Evaluation – ROC Curve & AUC
a) Plot the ROC curve.
b) Calculate the AUC score.